

Capacitive Quenching Measurement Circuit for Geiger-Mode Avalanche Photodiodes

S. J. Dimler, J. S. Ng, *Member, IEEE*, R. C. Tozer, *Senior Member, IEEE*, G. J. Rees, and J. P. R. David, *Senior Member, IEEE*

Abstract—Characterization of Geiger-mode avalanche photodiodes operated in gated mode requires fast rising-edge, well-defined over-bias pulses, and effective avalanche quenching. There has not been a suitable circuit that meets both criteria, thus, making systematic characterization and accurate comparison of these devices difficult. We present a capacitive quenching circuit (CQC) that satisfies both criteria and, thus, offers advantages over existing options such as a gated passive quenching circuit (G-PQC) and gated-mode operation without avalanche quenching. Test results using a commercial Si Geiger-mode avalanche photodiode, together with an experimental comparison between the CQC and the standard G-PQC circuit, are reported. The advantages of the CQC over the G-PQC circuit are demonstrated through comparisons of experimental over-bias pulses, distribution of avalanche current pulses, and dark count rate data. By each metric, the CQC is shown to yield results that are superior to those obtained using a standard G-PQC circuit.

Index Terms—Avalanche photodiode (APD), dark count, Geiger mode, impact ionization, quenching, single-photon detection.

I. INTRODUCTION

A SINGLE-PHOTON avalanche diode (SPAD) is an avalanche photodiode (APD) operated in Geiger mode. The SPAD is reversed biased beyond its avalanche breakdown voltage V_B so that a breakdown event may be initiated (hopefully) by a photon or (parasitically) by a thermally generated carrier. These devices have a range of applications where sensitivity to individual photons is required. Some examples are optical testing of CMOS [1], time-resolved photon counting [2], quantum key distribution [3], optical time-domain reflectometry [4], [5], and satellite laser ranging [6].

The rapidly growing avalanche current in a photon or dark-current-induced avalanche event must be quenched to avoid the possibility of damage to the SPAD. This can be achieved by using an external circuit, which can be either passive or active in nature giving rise to the terms passive and active quenching [7].

Passive quenching relies on a ballast resistor R_B , usually kilohms or tens of kilohms, in series with the SPAD. If an avalanche is initiated, the avalanche current produces a voltage drop across R_B , reducing the voltage across the SPAD to V_B [7] and leaving a residual avalanche current through the device. Since impact ionization is a random process, it is possible

for the magnitude of this residual avalanche current to spontaneously fall below a self-sustaining threshold, thus, quenching the avalanche completely. Complete quenching is more likely for smaller residual current so that a larger series resistor will, on average, quench the device sooner. When the avalanche current ceases, the potential across the SPAD rises again to that of the dc supply, and the device is again able to respond to a photon or thermally generated carrier. However, the simple passive quenching circuit suffers from a number of drawbacks. The ballast resistor, together with the SPAD capacitance, forms a low-pass filter with a time constant of the order of 100 ns to many microseconds. If the ballast resistor value is reduced to minimize the low-pass filtering, the residual current through the device is higher, and thus, quenching takes longer. High avalanche currents also encourage undesirable “after-pulsing” in which the avalanches are initiated by the release of trapped carriers in the SPAD.

Active quenching can provide more effective quenching than passive quenching by utilizing a circuit that detects the large avalanche current and lowers the applied bias to a value that will terminate the avalanche process. These circuits are far more complex in operation than their passive counterparts and, when constructed from discrete components, it can be difficult to achieve the desired high-speed operation and precise timing. The magnitude and duration of the potential used to quench the device will depend on the device properties [8]. Thus, for characterizing different SPAD designs, a very flexible active quenching circuit would be required.

The simplest mode of operation for a SPAD is the free-running mode, in which the SPAD is simply biased beyond its breakdown voltage using a dc supply. When breakdown occurs, the avalanche current can be quenched actively or passively. The free-running mode is useful for keeping the SPAD over-biased for the greatest time possible to detect weak photon fluxes, and is most suited for high-quality Si SPADs, which have a low dark count rate.

Devices such as InGaAsP/InP [10], [11], InGaAs/InP [12], [13], and InGaAs/InAlAs [14] SPADs, which detect photons with wavelength $>1\ \mu\text{m}$, have inherently higher dark count rates than Si SPADs due to the narrower bandgap absorber and growth defects. When the dark count rate is high, the time used to quench the large number of dark events will reduce the time available for photon detection. These devices are, therefore, operated in a mode known as “gated mode” where rectangular voltage pulses are used to over-bias the SPAD in coincidence with the expected photon arrival. A bias-tee is used to superimpose this series of rectangular pulses onto a dc potential, which is slightly below

Manuscript received January 14, 2007; revised June 26, 2007. This work was supported by the Engineering and Physical Sciences Research Council, U.K.

The authors are with the Department of Electronic and Electrical Engineering, University of Sheffield, Sheffield S1 3JD, U.K. (e-mail: s.dimler@shef.ac.uk; j.s.ng@shef.ac.uk; r.c.tozer@shef.ac.uk; g.rees@shef.ac.uk; j.p.david@shef.ac.uk).

Digital Object Identifier 10.1109/JSTQE.2007.903595

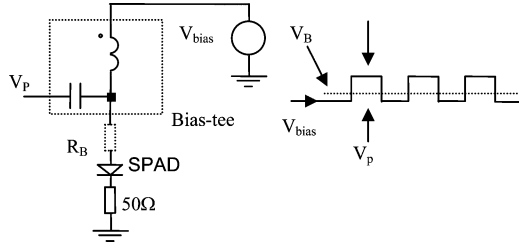


Fig. 1. Gated-passive quenching using a bias-tee.

the SPAD breakdown voltage, as depicted in Fig. 1. The pulse amplitude is chosen to bias the SPAD beyond breakdown for the duration of the pulse.

For very short pulses of a few nanoseconds, the gated mode can be used without quenching as the end of the pulse will bring the potential across the SPAD below breakdown before the avalanche current can grow to destructive proportions [5], [14]–[16]. However, longer over-bias pulses are essential for more comprehensive SPAD characterization and are useful for timing measurements [13].

To use longer over-bias pulses, some form of quenching is required. If passive quenching is used to give a gated passive quenching circuit (G-PQC), the low-pass filter formed by the SPAD capacitance and the ballast resistor R_B significantly modifies the shape of the applied over-bias pulse. Additionally, the bias-tee is a high-pass filter, which causes the over-bias pulse to reach a peak value and then fall for the rest of the pulse duration. This distortion becomes increasingly significant as the pulse duration is increased. Due to the random nature of avalanche breakdown events, the avalanche current pulses can vary in both shape and duration; hence, the total charge flowing through the APD varies, complicating the after-pulsing measurements. In essence, the lack of clean stable over-bias pulses and the presence of significant variations in avalanche current pulses make it difficult to interpret SPAD characterization data obtained using a G-PQC.

To use an active quenching circuit in gated mode, a fast recharge circuit can be utilized to improve the pulse shape of the applied over-bias. A typical fast recharge circuit attempts to charge the device capacitance quickly by switching the device to a low-impedance path [9]. However, this means that if the SPAD breaks down while recharge is underway, a large avalanche current can flow and the SPAD may not be protected. Thus, the recharge mechanism must be of sufficient duration to charge the capacitance associated with the device, but not so long that the breakdown during the recharge could damage the device [7], [9]. Additional checks can be incorporated to ensure that the device is biased correctly, and repeat the recharge if necessary [7]. Active quenching circuits usually perform best when they are designed for a specific device. This makes them unsuitable for characterization of a wide range of SPAD designs, whose junction capacitances, device area, and total depletion width may vary from device to device. Currently available active quenching circuits are designed for particular SPADs for which they have been optimized but may not have the flexibility to be modified for use with different devices.

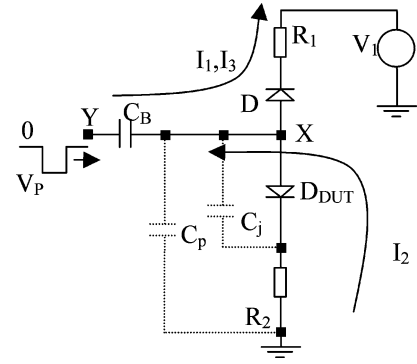


Fig. 2. CQC diagram.

Hence, there is a need for an alternative SPAD biasing/quenching circuit that is simple to use and construct, enabling consistent SPAD characterization for a wide range of SPAD and operating conditions. The circuit must deliver fast, clean, and stable over-bias pulses, and provide fast avalanche quenching. Some control over the total charge flowing through the device during breakdown would be an advantage because it offers the possibility of studying after-pulsing and may even suppress it.

In this paper, we describe a capacitive quenching circuit (CQC), which meets the requirements on over-bias pulse shape and quenching as mentioned above. Dark count rate measurements on a commercial Si SPAD using the developed circuit are presented and used to demonstrate the advantages of the CQC over the G-PQC.

II. CIRCUIT OPERATION

Fig. 2 shows a schematic circuit diagram for the CQC incorporating a device under test (DUT). V_1 is a negative dc bias, whose magnitude is slightly lower than the device breakdown voltage V_B . Pulses with a height of V_p (also negative) are ac-coupled to the SPAD via the biasing capacitor C_B . In the absence of a pulse, i.e., with zero volts at point Y, on initial application of the bias V_1 , a current I_1 will flow through C_B , D , and R_1 until the potential is at point $X \approx V_1$. There will also be a small current flowing through R_2 , the SPAD capacitance C_j , and parasitic capacitance C_p , as the potential at point X changes from 0 to V_1 .

When a pulse arrives, the potential at point Y falls by V_p , and this change in potential is at transmitted point X, with slight attenuation due to capacitive division. The change in potential at X demands a change in charge in the diode capacitance $C_j + C_p$, and after a short time constant, the point X will settle at $V_X = V_1 + V_p \frac{C_B}{C_B + C_j + C_p}$, which, with appropriate choice of V_1 and V_p , will bias the SPAD beyond breakdown.

If no breakdown occurs during the applied pulse, the potential at point X will return to V_1 at the end of the pulse. If an avalanche breakdown occurs, current I_2 will be conducted by the SPAD until the magnitude of the potential at point X becomes approximately V_B , the point at which the SPAD is no longer over-biased and ceases to conduct. Quenching the SPAD to its breakdown voltage is similar to the behavior of resistive

quenching, but in this circuit, the quenching device is a capacitor. Thus, instead of limiting the avalanche current and hoping for a spontaneous termination, this circuit controls the total charge that can flow in the avalanche, and termination is guaranteed. During the avalanche breakdown, the current I_2 will generate a voltage across R_2 (in series with the DUT) that is similar in appearance to the output of a passive quenching circuit. R_2 must be sufficiently large to generate an avalanche pulse that is detectable. We have used $50\ \Omega$ because it satisfies the detectability condition and provides matched source impedance to a $50\ \Omega$ cable, a facility that is necessary for cryogenic measurements where the differential amplifier would be remote from the DUT. The negative voltage supplies used in the configuration shown in Fig. 2 result in an avalanche pulse that is compatible with widely available discriminators that have negative thresholds designed for photocathodes.

If breakdown has not occurred by the end of the pulse, the potential at point Y returns to zero, and the potential at point X returns to the original dc value of $\sim V_1$. If breakdown does occur within the pulse, the charge lost from the capacitance at node X to the avalanche must be replaced by current I_3 flowing through diode D_1 . The time taken for the potential at X to return to $\sim V_1$ is determined by the time constant formed by R_1 and the capacitors C_B , C_p , and C_j . R_1 protects the DUT from an accidental application of excessive V_1 . If V_1 is accidentally set at a level that biases the DUT above its breakdown voltage, R_1 is the only current limit in the external circuit. In applications where a high gate pulse repetition rate is needed, e.g., in the study of after-pulsing effects, R_1 can be reduced in value or removed completely. This gives a fast settling time but reduces or eliminates the protection offered by R_1 against accidental overbias.

The maximum gate pulse repetition frequency is limited by the recharge time, which, in turn, is dependant on the size of C_B . With $C_B = 27\ \text{pF}$, the time for 99% recharge will be approximately 6 ns. We have operated the circuit at 50 MHz, the maximum frequency of the pulse generator used. Fast recharge was not an issue for the measurements that we required; thus, $1\ \text{k}\Omega$ was used for R_1 to add protection from excess dc but leading to a recharge time constant, when breakdown occurs, of $\tau \approx R_1(C_B + C_p + C_j)$. The quenching time is dominated by the time for $C_B + C_p + C_j$ to discharge via the series resistance of the DUT; R_2 has little effect on it.

III. EXPERIMENTAL SETUP AND MEASUREMENT CIRCUIT

The circuit shown in Fig. 2 was employed in an experimental setup for dark count rate measurements, which is depicted schematically in Fig. 3. A PerkinElmer C30902S Si SPAD was used as the DUT. In order to measure the dark count rate of the DUT, the pulse shape across the DUT, and the avalanche pulse shape, the CQC was incorporated into a measurement circuit, as shown in Fig. 4. In the experimental setup, the dc bias was provided by a Keithley 6517 Electrometer, and the pulses were provided by an Agilent Technologies 81101A pulse-pattern generator with a 5 ns 10%–90% minimum rise time. The avalanche pulses (measured across the resistor R_2) and

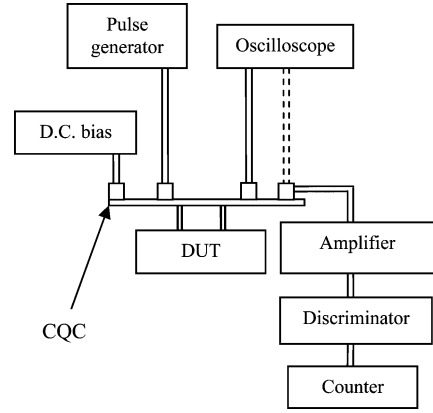


Fig. 3. Schematic diagram of the dark count rate measurement setup.

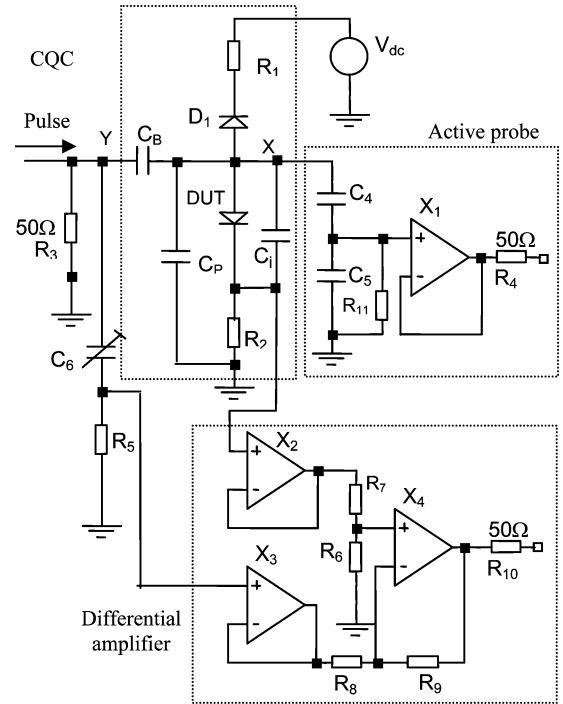


Fig. 4. Diagram of the dark count rate measurement circuit, which incorporates the CQC, the transient cancellation components, and the active probe.

the over-bias pulses (measured via an active probe) were monitored using an Agilent Technologies 54642A oscilloscope with 500-MHz bandwidth. An Ortec 583 constant-fraction discriminator was used to detect the avalanche events, and provided corresponding output NIM pulses fed to a Canberra 512 counter. The DUT was maintained at a constant temperature of 300 K using a thermoelectric cooler and a Keithley 2510 temperature controller.

In addition to monitoring various voltage signals of interest in the CQC, the measurement circuit shown in Fig. 4 includes transient cancellation features [7] to overcome transient voltage spikes across the resistor R_2 , which can lead to false counts of avalanche breakdowns. These voltage spikes are created by the current that passes through the capacitance C_j associated with the DUT when the over-bias pulse is applied. In the measurement

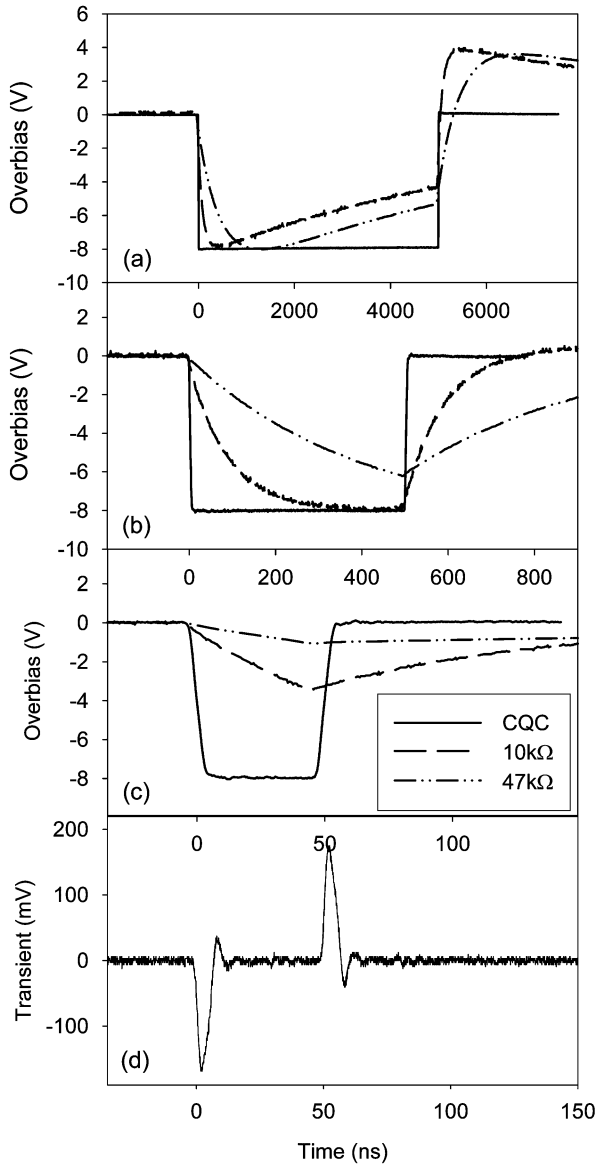


Fig. 5. Comparison of over-bias pulses, as measured across the DUT, using the CQC (solid lines) and the G-PQC (dashed and dashed-dotted lines for ballast resistor of 47 and 10 k Ω , respectively). Pulse durations of (a) 5 μ s, (b) 500 ns, and (c) 50 ns are shown, while (d) shows the transient across R_2 .

circuit, the over-bias pulses from the pulse generator were also applied to capacitor C_6 , resulting in voltage spikes across R_5 . C_6 was then adjusted so that the magnitude of the voltage spikes across R_5 is similar to that of R_2 . A differential amplifier formed by high-speed operational amplifiers X_2 , X_3 , and X_4 then subtracted the transient across R_5 from the output across R_2 . Fig. 5(d) shows the transient across R_2 resulting from the application of an 8 V pulse, with 5 ns 10%–90% rise time, to the DUT. The resulting transient of approximately 150 mV is reduced to <10 mV by the differential amplifier.

An additional benefit of using a differential amplifier is that the polarity of the output pulse from X_4 can be reversed by interchanging the connections made to the noninverting inputs of X_2 and X_3 . Thus, D_1 and DUT can be reversed in polarity

so that the cathodes are connected together at point X , allowing the circuit to be used with a positive dc bias while still providing a negative avalanche pulse simply by swapping the inputs to the differential amplifier. This provides additional flexibility, especially when one of the leads of the DUT is connected internally to the device package where it is desirable to connect this node to R_2 so that the capacitance between the package and the ground does not add to C_p .

To monitor the over-bias pulses applied to the DUT, the measurement circuit includes an active probe, as shown in Fig. 4. The probe is formed by the high-speed FET operational-amplifier X_1 , and the capacitive potential divider created by C_4 and C_5 . R_{11} is a 100 M Ω resistor, which provides a path for the input bias current of X_1 . The values for C_4 and C_5 are chosen to provide suitable potential division such that the maximum input voltage and slew rate of X_1 are not exceeded, and to minimize additional capacitive loading at point X . The addition of R_{11} to the capacitive divider results in a high-pass filter with a time constant of several milliseconds. A 1 ms time constant would cause a 1% fall in measured pulse height in a period of 10 μ s.

The outputs of the active probe and the differential amplifier are connected via 50 Ω terminations to the oscilloscope or discriminator, as appropriate. Using transient cancellation features and active probe identical to those shown in Fig. 4, a separate measurement circuit for the G-PQC was constructed for subsequent measurements.

IV. RESULTS

A dc bias of 231.8 V, 100 mV below the level at which small (<50 μ A) self-quenching avalanche events were observed, was applied to the SPAD using both the CQC and a conventional G-PQC. Fig. 5 compares the over-bias pulses measured across the Si SPAD obtained using the two circuits. The data shown were obtained using pulse widths of 5 μ s, 500 ns, and 50 ns, as measured at the output of the pulse generator between the 50% level of the rising and falling edges. For the G-PQC, the ballast resistor values of 47 and 10 k Ω were tested. The over-bias pulses were significantly distorted when the G-PQC was used. The high-pass filtering of the bias-tee caused the pulse height to decrease with time, which is noticeable for the 5 μ s pulse width. For the shorter pulses, the over-bias pulses across the SPAD were severely affected by the low-pass filtering of the SPAD capacitance in combination with the ballast resistor. A lower value ballast resistor improves the pulse shape but degrades the quenching performance, as will be discussed later.

In contrast, Fig. 5 shows that the CQC applies clean and stable pulses across the SPAD. The pulse across the DUT is the difference between the applied pulse and the potential across R_2 ; for the 8 V applied pulse, the peak of the transient across R_2 is only 1.88% of the applied pulse height, and falls to zero during the flat portion of the over-bias pulse. The rise and fall times of the pulses across the SPAD were determined by the pulse generator used; there was no observable degradation introduced by the CQC for any of the pulse widths used. The well-defined over-bias pulses enable consistent measurements of avalanche breakdown probability and accurate analyses since

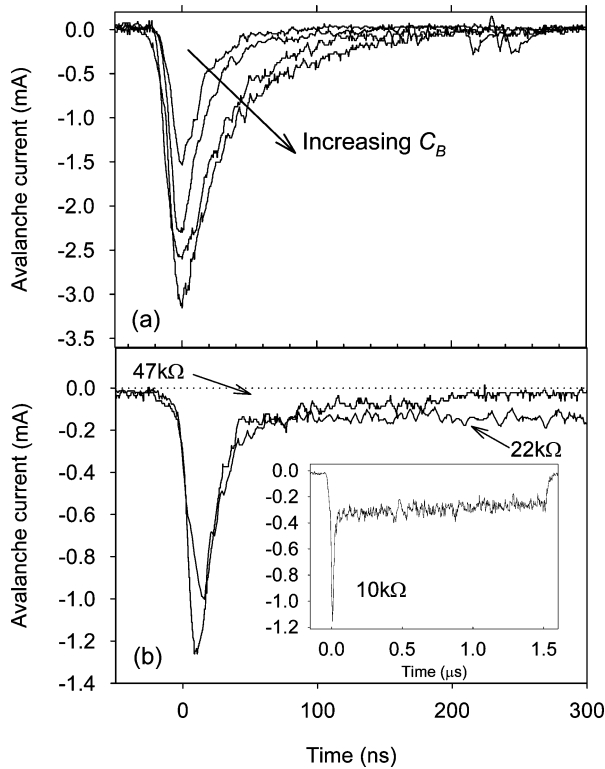


Fig. 6. Sample avalanche current pulses obtained with 4-V over-bias pulses applied to the Si SPAD, using (a) the CQC, with C_B of 5.6, 15, 33, and 47 pF, and (b) the G-PQC, with ballast resistor of 47, 22, and 10 kΩ (inset).

they maintain a constant over-bias voltage over their duration. With CQC, the pulse height decay is due only to leakage in the SPAD, blocking diode D_1 , and the capacitor C_B . In the measurements with the Si SPAD, we observed no noticeable pulse height decay due to these factors.

Using the setup and the CQC shown in Figs. 3 and 4, respectively, the avalanche current of each breakdown event of the Si SPAD was recorded for over-bias pulse height of 4 V and duration of 1 μ s. Fig. 6(a) compares the samples of the avalanche current pulses obtained using the CQC, with C_B of 5.6, 15, 33, and 47 pF. Increasing C_B gives avalanche pulses of greater magnitude, which can be used to give a stronger signal so that transient cancellation is less critical, or even unnecessary. However, using a smaller C_B in the CQC with voltage transient cancellation is still advantageous because it gives faster quenching.

The Si SPAD avalanche current pulses for the same over-bias pulses (4-V height and 2 μ s duration) were also recorded for the G-PQC. Ballast resistors of 47, 22, and 10 kΩ were tested, and the data are compared in Fig. 6(b). Due to the distorted over-bias pulse shapes, as shown in Fig. 5, only avalanche events beginning within a 250 ns time window were accepted. The 250 ns window was positioned on the flattest and highest part of the 2 μ s over-bias pulse. This is consistent with the common practice of using a G-PQC with injected photons timed to arrive during the flattest part of the gate.

It can be observed from Fig. 6(b) that using a large ballast resistor provides more effective avalanche quenching. Compared to a smaller ballast resistor, a larger ballast resistor results in a

smaller avalanche current flowing through the SPAD, increasing the probability of the self-sustaining avalanche terminating before the over-bias pulse is over. It is desirable to have short self-sustaining avalanche events in order to minimize the number of carriers trapped in the SPAD, which will encourage after-pulsing. Therefore, when using the G-PQC circuit, there is a compromise between clean over-bias pulses and effective avalanche quenching. A typical avalanche pulse measured when using the 10 kΩ ballast resistor, as shown inset in Fig. 6(b), exhibited self-sustaining avalanche until the over-bias pulse returned to zero. The magnitude of the avalanche current, during the self-sustaining avalanche decreased slightly with time, which is expected due to the decreasing over-bias pulse height shown in Fig. 5(a).

The commercial discriminator/counter combination used in our system is active throughout the gate period. Therefore, in the measurements using the G-PQC gated circuit, the discriminator threshold has to be set to avoid the level of the sustained avalanche current where noise could lead to false counts. However, this results in missing counts from avalanche events occurring when the over-bias magnitude and the avalanche currents are small. The clean avalanche current pulses obtained using the CQC do not suffer from this problem because, following an avalanche, the device voltage remains too low for further activity to occur until the end of the gating pulse, thus, allowing accurate detection of avalanche breakdown events and the time at which they occur.

For each avalanche breakdown event, the corresponding avalanche current pulse varies, and this has significance in measurements of avalanche event count rate and the associated timing information. The timing information of interest is the time elapsed between the arrival of an injected photon and the detection of a growing avalanche current due to the excitation. Variation in this time is the timing jitter of the avalanche breakdown process. The low value of R_2 , in combination with the capacitance of the SPAD, results in a small time constant that is not evident in the measured waveforms due to the bandwidth of X_2 , X_3 , and X_4 .

In order to study the variation in the shape of avalanche current pulses for given experimental conditions, the mean avalanche current pulse and its standard deviation, $\langle I(t) \rangle$ and $\sigma(t)$, respectively, were calculated from 1000 avalanche event samples. The $\langle I(t) \rangle$ and $\langle I(t) \rangle \pm \sigma(t)$ for the Si SPAD with 8 V over-bias pulses applied, obtained using the CQC and the G-PQC (with 10 kΩ ballast resistor) measurement circuits, are plotted in Fig. 7. It should be noted that for the G-PQC circuit, only the avalanche pulses beginning within the 250 ns time window, where over-bias pulse height varies the least, were accepted as described above. Despite this, the avalanche current pulses obtained using the CQC, as shown in Fig. 7(a), show smaller standard deviation of 8.6% of the peak current *versus* 14.3% variation of the peak for the G-PQC circuit.

The inset graph in Fig. 7(b) plots two examples of the avalanche current pulses obtained using the G-PQC with a 10 kΩ ballast resistor to show the range in the avalanche duration. The variation is greater than 100 ns, and would be larger if the over-bias was not decreasing in magnitude due to the

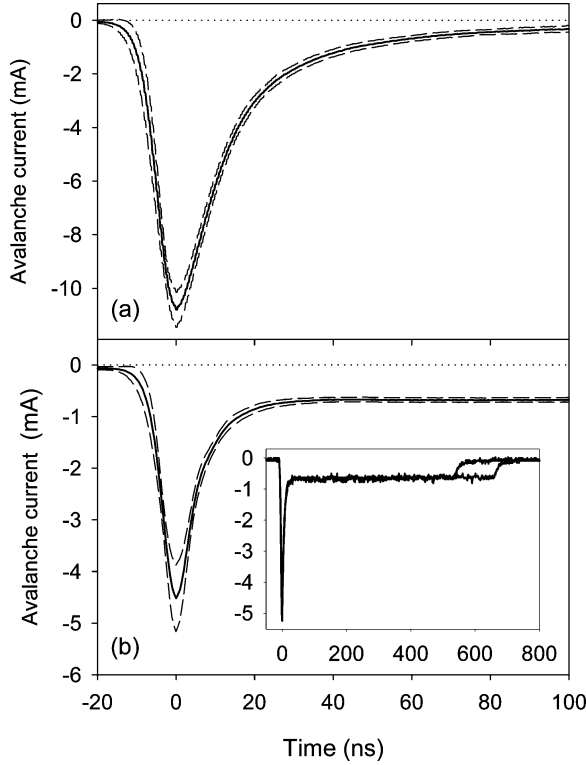


Fig. 7. $\langle I(t) \rangle$ (solid lines) and $\langle I(t) \rangle \pm \sigma(t)$ (dashed lines) of the Si SPAD with 8 V over-bias pulses applied. The data for (a) the CQC (with a 33 pF biasing capacitor) and (b) the G-PQC circuit (with a 10 kΩ ballast resistor) are shown. The inset in (b) illustrates the variation in sustained avalanche duration for the G-PQC.

high-pass filtering of the bias-tee. This variation in avalanche duration results in a variation in the total amount of charge flowing through the SPAD in each avalanche event. Such variation is reduced when the CQC was used, providing a possibility of experimental studies of relationship between the after-pulsing problem and the total amount of charge flowing through the SPAD simply by changing the capacitor C_B . These studies may even be possible in the absence of light, thus, providing another tool to researchers in addition to the current approaches for after-pulse investigation [10], as charge can be varied independently of over-bias magnitude.

We also measured the dark count rate r_d of the Perkin Elmer Si SPAD as a function of over-bias pulse width τ , using the two measurement circuits. The dc bias and the pulse repetition frequency f were kept constant at 231.8 V and 10 kHz, respectively. The voltage pulses from the pulse generator were adjusted to give a peak rectangular pulse height of 8 V as measured on the DUT. The experimental dark count rate *versus* the over-bias pulse-width characteristics obtained using the CQC and the G-PQC circuit (ballast resistors of 47 and 10 kΩ were tested) are compared in Fig. 8.

Assuming that the events initiating dark current have Poisson characteristics, the probability of n events leading to breakdown occurs during an over-bias pulse of $p(n) = \frac{e^{-\mu} \mu^n}{n!}$, where $\mu = \lambda\tau$ is the mean number of such events in time τ , and λ is the dark count rate in free-running mode with fast quenching. Since only one such event is counted, the probability of a count is 1 minus

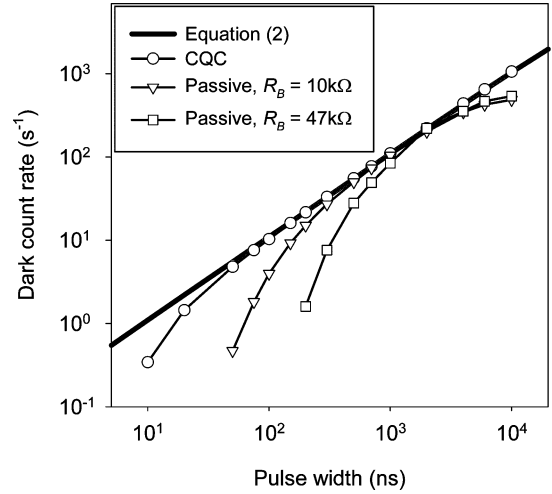


Fig. 8. Dark count rate *versus* over-bias pulse width measured using the CQC and the G-PQC circuit (symbols). A least-squares fit to the CQC data is also shown (grey line).

the probability of no dark events, $1 - p(0) = 1 - \frac{e^{-\mu} \mu^0}{0!} = 1 - e^{-\mu} = 1 - \exp(-\lambda\tau)$.

The dark count rate is just the product of this probability and the pulse repetition frequency

$$r_d = f(1 - \exp(-\lambda\tau)). \quad (1)$$

When the product $\lambda\tau$ is small, (1) becomes

$$r_d \approx f\lambda\tau. \quad (2)$$

Using the CQC, the experimental dark count rate increases proportionally with the over-bias pulse width for moderately wide over-bias pulses (~ 50 ns to ~ 10 μ s), as expected from (2). A least-squares fit to the data indicates $\lambda = 11$ 000 counts per second. As the over-bias pulse width decreases to below 50 ns, the rise and fall times, due to the pulse generator used, become significant compared to the over-bias pulse width. This gives rise to lower than expected dark count rate, resulting in deviation from the linear relationship between the dark count rate and the over-bias pulse width.

It is interesting to note that if ~ 7 ns is subtracted from the pulse duration given by the pulse generator, corresponding to a pulse duration measured at the 90% level, then all the data points for the CQC perfectly follow (2). This suggests that this is the effective pulse duration as far as the SPAD is concerned. This is perhaps not surprising as both breakdown probability and dark carrier generation are strongly dependent on the over-bias voltage.

It can be observed from Fig. 8 that the dark count rate measured using the G-PQC circuit was generally lower than that measured using the CQC. The disagreement between the data from the different measurement circuits grows as the over-bias pulse width decreases, due to increasing pulse shape distortion in the G-PQC. Furthermore, at longer pulse widths, Fig. 8 shows that dark count rate tends to decrease as pulse width increases when the G-PQC is used. This behavior is attributed to the over-bias pulse height decreasing with time. Using a smaller ballast

resistor helps to reduce the disagreement between the data obtained using the G-PQC circuit and the CQC. This is, however, at the expense of less-effective avalanche quenching.

V. CONCLUSION

We have presented a novel circuit CQC for gated-mode SPAD characterization, which simultaneously provides clean stable over-bias pulses and effective quenching. The circuit effectively increases the usable over-bias pulse width commonly seen with gated-mode measurements by three orders of magnitude (from <10 ns to >10 μ s). This provides the means, as demonstrated, to obtain the mean dark count rate for a given SPAD. The circuit gives efficacious repeatable avalanche quenching with avalanche current pulse magnitude and duration dependent on the biasing capacitor. This also allows the charge passing through the SPAD to be varied independently of over-bias amplitude, facilitating systematic after-pulse investigation. The circuit allows the pulse generator to be ac-coupled, which may allow greater flexibility when choosing or designing high-speed pulse generators.

ACKNOWLEDGMENT

The authors wish to thank Prof. G. Buller of Heriot-Watt University, Edinburgh, for discussions concerning passive quenching circuits and device characterization.

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S. J. Dimler received the M.Eng. degree in electronic and electrical engineering from the University of Sheffield, Sheffield, U.K., in 2004, where he is currently working toward the Ph.D. degree in electronic and electrical engineering.

His current research interests include developing circuits and measurement setups for characterization of photodetectors.

J. S. Ng (M'99) received the B.Eng. and Ph.D. degrees from the Department of Electronic and Electrical Engineering, University of Sheffield, Sheffield, U.K., in 1999 and 2003, respectively.

From 2003 to 2006, she was with the University of Sheffield, where she became a Royal Society University Research Fellow in October 2006, where she is responsible for characterization within the National Centre for III–V Technologies. Her research interests include avalanche photodiode, Geiger mode avalanche photodiode, and material characterization.

R. C. Tozer (SM'98) received the B.Eng., M.Eng., and Ph.D. degrees from the University of Sheffield, Sheffield, U.K., in 1970, 1972, and 1975, respectively.

He was a Postdoctoral Researcher with the University of Sheffield, after which he became a Lecturer with the University of Essex, Colchester, Essex, where he carried on research on active sound cancellation. He returned to the University of Sheffield as a Lecturer in 1980, where he currently teaches analogue circuit design. His research interests include application of analogue circuits to a wide range of experimental and instrumental problems, particularly APD noise measurement systems and novel excitation modes for fluorescent lamps.

Dr. Tozer is a Chartered Engineer and a member of the Institution of Electrical Engineers.

G. J. Rees received the B.A. degree in physics from Oxford University, Oxford, U.K., in 1966, and the Ph.D. degree in theoretical physics from Bristol University, Bristol, U.K., in 1969.

He has been a Lecturer of mathematics at Imperial College, London, U.K. In 1971, he joined the Plessey Company's Device Research Laboratory, Caswell, Swansea, U.K., where he was the Head of the Theoretical Technology Group. He was a visiting fellow at Lund, Sweden, and the Clarendon Laboratory, Oxford. In 1991, he joined the Department of Electronic and Electrical Engineering, University of Sheffield, Sheffield, U.K., where he was the Chair in 1999 and the Head of Department in 2001, and is now an Emeritus Professor. His current research interests include avalanche photodiodes and modeling of semiconductor device physics.

Prof. Rees was awarded a one-year Royal Society Exchange Fellowship at the Università delle Scienze, Rome, Italy.

J. P. R. David (SM'96) received the B.Eng. and Ph.D. degrees in electronic engineering from the University of Sheffield, Sheffield, U.K.

He joined the Central Facility for III–V Semiconductors, University of Sheffield, in 1985, where he was responsible for the characterization activity. In 2001, he joined Marconi Optical Components (now Bookham Technologies), Devon, U.K., before returning to a Faculty Position in the University of Sheffield in September 2002. He has authored or coauthored more than 250 papers published in various journals and conference proceedings, largely in the areas of III–V characterisation, impact ionization, and avalanche photodiodes.